

ARNAB JANA
Data Scientist | Machine Learning Engineer | Quantitative Research demo demo demo demo
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GitHub: github.com/Arnab025-wq/Arnab_Jana
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Portfolio: arnabjana.github.io

EDUCATION
Indian Institute of Technology Madras
BS in Data Science and Applications | 2026–2030
CGPA: 9.4/10 (demo)
Indian Institute of Banking & Finance
Diploma in Banking & Finance (DBF) | Ongoing

EXPERIENCE
Machine Learning Research Intern
Research Lab / AI Startup | Bengaluru / Remote
May 2028 – Aug 2028 (demo)

- Developed a **probabilistic forecasting pipeline** for high-frequency financial data using Python, NumPy, Pandas and PyTorch.
- Improved out-of-sample forecasting error by **23%** over the baseline statistical model.
- Presented findings to a research team of 8 engineers and researchers.

Data Science Intern
FinTech Company | Mumbai
May 2027 – Jul 2027 (demo)

- Built a credit-risk modelling pipeline using **XGBoost, logistic regression and calibrated probability estimates**.
- Developed automated model monitoring dashboard for drift and performance analysis.

RESEARCH
Statistical Learning for Financial Time Series
Independent Research Project | 2027–28

- Investigated the relationship between statistical dependence, volatility clustering and predictive performance in financial time series.
- Compared ARIMA, GARCH, Random Forest and Transformer-based approaches.

Remote Sensing & Earth Observation
Research Project | 2027

SELECTED PROJECTS
Market Regime Detection using Machine Learning
Python · PyTorch · Pandas · NumPy · Statistics

- Built Hidden Markov Model, Random Forest and LSTM approaches for identifying market regimes.
- Evaluated models using **walk-forward validation rather than random train/test splitting**.
- GitHub: github.com/Arnab025-wq/market-regime

Flood Risk Mapping using Remote Sensing
Python · GIS · Raster Processing · Machine Learning

- Developed geospatial pipeline combining rainfall, elevation, land-cover and satellite-derived features.
- Generated flood susceptibility maps for a flood-prone region of West Bengal.
- Compared logistic regression, Random Forest and gradient boosting models.
- Designed methodology with emphasis on interpretability and reproducibility.

TECHNICAL SKILLS
Languages: Python, C++, C, SQL, R
Machine Learning:
Regression, Classification, Clustering, Ensemble Methods, Neural Networks, Time Series, NLP, Computer Vision
Mathematics & Statistics:
Linear Algebra, Calculus, Probability, Statistical Inference, Optimization, Hypothesis Testing, Bayesian Methods
Tools:
Git/GitHub, VS Code, Jupyter, Linux, MATLAB

OPEN SOURCE & RESEARCH

- Maintainer/contributor to open-source Python projects related to data science.
- Published reproducible implementations of statistical and machine-learning research.
- Built documentation and testing infrastructure for research code.
- Regularly reproduce results from academic papers before extending methodologies.

LEADERSHIP & ACTIVITIES Ramanujan Society of Research, IIT Madras
Research MemberData Science / AI Community
Organized workshops on Python, statistics and machine learning for undergraduate students.
Hackathon Team Lead
Led a 4-member interdisciplinary team developing an AI-based climate-resilience solution.